

Topic Selection

Location Privacy Challenges and Solutions : Location Transformation, MAC Rotation, and other solutions

Abstract

In this world where more and more services require location to run, the anonymization of data and the privacy of users is a must. From car hijack to house burglary or even military attacks on predictable roads, there are a lot of different attacks possible with just the ability to know, process and manipulate location data. Furthermore, many actors are in the circuit, needless to say the receiver, who needs to use the service that requires his location but also the services. However there are plenty of third-parties involved and sometimes the service may use the location of the receiver in a dishonest manner like selling to other 3rd-parties.

This project aims to discuss about potential solution both from the user and from the server/protocol perspective. This paper is about creating a foundation for future research and helping clarify how the different objectives can or cannot be accomplished. It will also cover the base of what a receiver should do to stop being spy on, and new protocols that ensure that the receiver remains anonymous.